

Social Networks: Assignment #3

Due on Thursday, January 4, 2026

Dr. Masoud Asadpour 15:00am

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Contents

1		4
(a)	Using the standard graph reduction algorithm based on positive connected components, construct the reduced graph G' . Clearly specify the vertices of G' (i.e., the supernodes), listing the original nodes contained in each supernode.	4
(b)	By analyzing the structure of the reduced graph G' , determine whether the original network G satisfies the property of strong structural balance. Provide your argument solely based on the examination of cycles present in the graph.	4

List of Figures

Question 1

Graph Reduction in Large and Irregular Signed Networks

(a) Using the standard graph reduction algorithm based on positive connected components, construct the reduced graph G' . Clearly specify the vertices of G' (i.e., the supernodes), listing the original nodes contained in each supernode.

After running the graph reduction algorithm we reach 5 super-nodes listed below:

- super-node A containing nodes {6, 7, 8}
- super-node B containing nodes {1, 2, 3, 4, 5}
- super-node C containing nodes {14, 15, 16, 17, 18}
- super-node D containing nodes {10, 11, 12, 13}
- super-node E containing node {9}

The edges of the reduced graph are all annotated with $-$ sign and they are $\{AB, BC, CD, DE, EA, BD\}$

(b) By analyzing the structure of the reduced graph G' , determine whether the original network G satisfies the property of strong structural balance. Provide your argument solely based on the examination of cycles present in the graph.

We know that our graph is not balanced if we find a cycle in the reduced graph that has an odd number of edges the cycle $ABCDEA$ has 5 edges thus this graph is not balanced

References

- [1] ChatGPT, *Chat session used for assignment guidance*, <https://chatgpt.com/share/69518d24-9800-800d-8fe8-185a102b5f10>, accessed December 28, 2025.